

Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

1. Introduction

2 hrs · asked in **all 22 papers** · 3 syllabus sub-topics · no numerical component.

Read this first:

- Shortest chapter in the syllabus, but **every single paper opens with it**, usually as Q1 for 4 to 6 marks. Cheapest marks in the subject.
- Only **two** questions are ever asked, in rotation:
 - **Narrate** the evolution (1G → 5G) **TOP 10/22**,
 - **Compare** the generations in terms of technology advancement **TOP 9/22**.
- Both are answered from **one** table (§1.3). Learn that table and the chapter is finished.
- Two small riders recur: **forward and reverse channel** **HOT 4/22** and **trends beyond 3G / 5G** **HOT 4/22**. 2 marks each, one line each.
- **Do not write history as prose**. The examiner wants generation, access technique, modulation, data rate, example system. Dates alone score badly.

1.1 Evolution of mobile radio communications

Briefly explain the evolution of wireless / mobile communications **4 5 6**

TOP 10/22 70 Ma **6**, 72 Ma **6**, 71 Bh **4**, 74 Ma **4**, 75 Bh **4**, 71 Ma **4**, 73 Bh **4**, 76 Bh **6**, 78 Ch **4**, 81 Bh **2+4** · **wording changes, the answer does not**

Milestones, before cellular

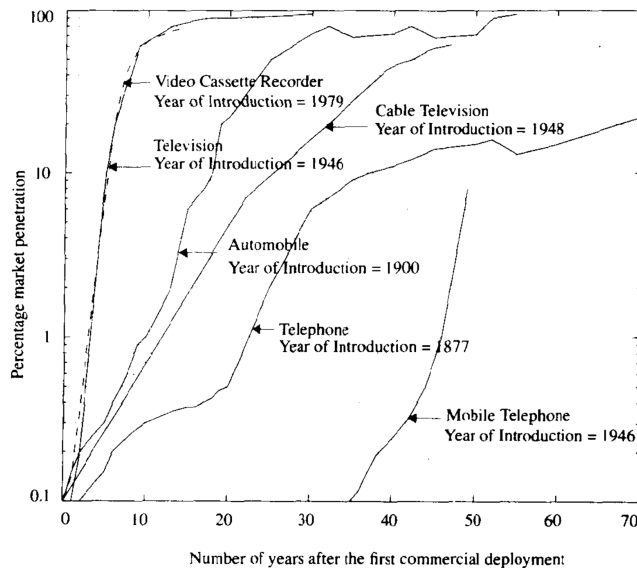
- **1877**: first wire-line telephone in commercial use.
- **1897**: Marconi demonstrates radio to a tug 18 miles at sea.
- Until **1934**: mobile radio = AM, used only by police and public bodies.
- **1935**: Armstrong invents **FM** → becomes the mobile standard for the next 40 years.
- **1946**: first public mobile telephone service, 25 US cities.
 - Single high-power transmitter, tall tower, range over 50 km.
 - Half-duplex push-to-talk, 120 kHz of RF for 3 kHz of speech.
- **1950**: US regulator doubles the channels, no new spectrum. Bandwidth per channel halved to 60 kHz, later 30 kHz.
- **Late 1950s / 60s**: **IMTS** (Improved Mobile Telephone Service) adds full duplex, auto-dial, auto-trunking.

Milestones, cellular era

- **1960s**: AT&T Bell Labs develop cellular radiotelephony theory.
- **1968**: AT&T proposes a cellular system to the US regulator.
- **1979**: NTT, Japan. **World's first commercial cellular system**.
- **1983**: **AMPS** starts in the USA. The regulator allocates 40 MHz in the 800 MHz band = 666 duplex channels of 30 kHz each way.
- **1990 onwards**: 2G digital systems, GSM and DCS 1800.
- **1995**: 3G standardisation starts, IMT-2000 / UMTS.
- **2000s**: 3G deployed, then 4G LTE.
- **2020**: 5G commercial.

The bottleneck that forced the cellular idea

- Bell Mobile Phone, New York, **1976**:
 - market of 10,000,000 people, **12 channels**, **543 paying customers**, waiting list over **3,700**.
- Spectrum is fixed by the regulator → capacity could not grow by adding transmitters.
- **Only fix left: reuse the same frequencies inside the same city.** That is the cellular concept (Ch 2).



Growth of mobile telephony against other consumer technologies. Note the 35-year flat start. Rappaport Fig 1.1

Worldwide market penetration + Fig

- Mobile telephony was introduced in **1946** but sat below 0.1 % penetration for **35 years**, until cellular arrived.
- Once cellular started it grew **faster than any earlier consumer technology**: US cellular users 25,000 (1984) → 16 million (1994), growth over 50 % per year.
- **This is the point of the figure**: the curve is flat while the technology is pre-cellular, then turns almost vertical.

Answer skeleton 4 6

- 1. Pre-cellular era (1946–1979)**: one high-power transmitter, no reuse, a dozen channels per city → blocked calls, waiting lists.
- 2. Cellular idea (1968–1979)**: many low-power cells + frequency reuse → capacity grows w/o new spectrum.
- 3. 1G (1983)**: analog, FDMA/FDD, AMPS.
- 4. 2G (1990s)**: digital, TDMA or CDMA, GSM / IS-95.
- 5. 2.5G**: packet data bolted on, GPRS / EDGE.
- 6. 3G (2000s)**: IP + broadband, W-CDMA / cdma2000.
- 7. 4G (2010s)**: all-IP, OFDMA + MIMO, LTE.
- 8. 5G (2020)**: mmWave, massive MIMO, IoT.

For a 6-mark answer add one line of *why* each step happened.

The examiner is looking for the driver, not just the date.

1.2 Simplex, duplex, FDD and TDD

Define forward and reverse channel 2

HOT 4/22 79 Bh, 79 Ch, 80 Bh, 80 Ch · always 2 marks, always paired with a 5G part

What do you mean by duplexing? Explain its different types 2+4

PIN 1/22 81 Ba

Three transmission modes

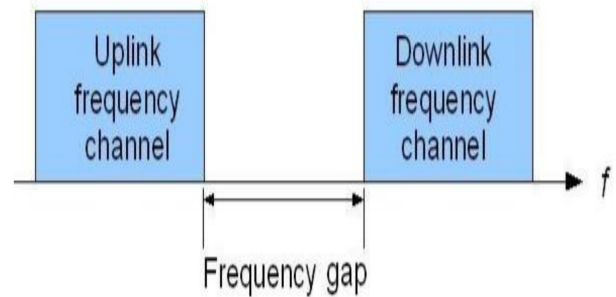
- **Simplex**: one direction only. Eg paging, broadcast radio.
- **Half duplex**: both directions, **not at once**. One radio channel, push-to-talk. Eg walkie-talkie.
- **Full duplex**: both directions **simultaneously**. Needs two channels, or one channel split in time.

Duplexing = how full duplex is achieved

- **FDD**, frequency division duplexing:
 - Two **separate frequencies**, one per direction.
 - Separated by a **frequency gap** (guard band) so the handset's own transmitter does not swamp its receiver.
 - Eg AMPS and GSM: forward and reverse bands 45 MHz apart.
 - Needs a **duplexer** (costly RF filter) in every handset.
- **TDD**, time division duplexing:
 - **One frequency**, shared in **time**. Part of each frame carries BS → MS, the rest MS → BS.
 - No duplexer needed → cheaper, smaller handset.
 - **Only possible with digital transmission**, and only over short ranges, because propagation delay eats the guard time.
 - Sensitive to timing. Eg DECT cordless, WiMAX, 5G NR.

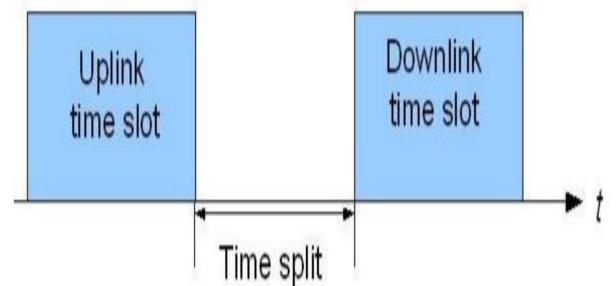
Forward and reverse channel (the 2-mark definition)

- **Forward channel** = **downlink** = base station → mobile.
- **Reverse channel** = **uplink** = mobile → base station.
- The air interface defines **four** physical channels, not two:
 - forward voice / data channel (FVC), reverse voice / data channel (RVC),
 - forward control channel (FCC), reverse control channel (RCC).
- Control channels carry setup, paging and service requests. About **5 %** of the total channels.
- **Reverse link is the harder one to design**: the handset has a small battery and a poor antenna, so the reverse link limits cell radius (see the Ch 3 link budget numericals).



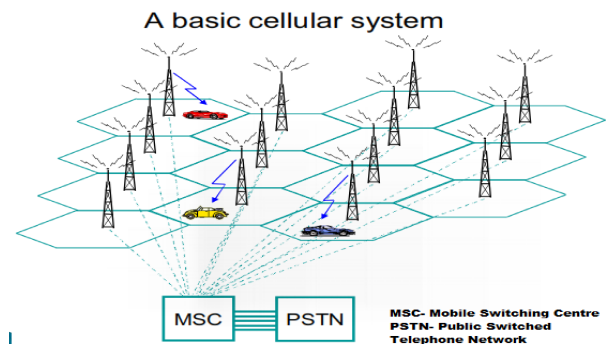
(a) FDD

FDD: two frequencies, permanent guard gap.



(b) TDD

TDD: one frequency, direction toggled in time.



A basic cellular system. Every BS reaches the MSC, and the MSC reaches the PSTN.

1.3 Generations of cellular radio: 1G to 5G

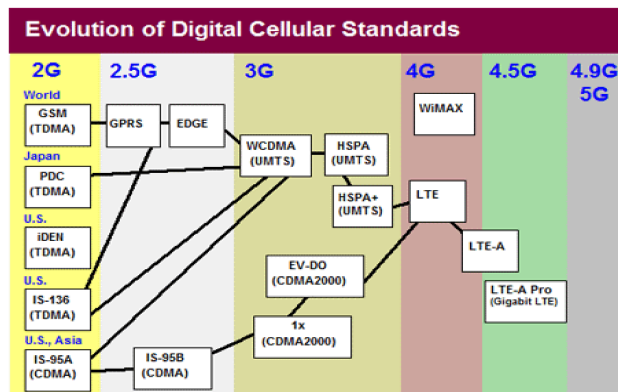
Compare the generations of mobile communication standards in terms of technology advancement

4 5 6

TOP 9/22 72 Ash 4, 75 Bh 4, 74 Bh 6, 80 Ba, 81 Ba 4, 80 Bh 4, 70 Bh 6, 77 Ch 5, 73 Ma 4 · this one table answers every variant, 1G Vs 2G, 3G Vs 4G, 4G Vs 5G

Gen	Era	Signal + access	Modulation	Peak data rate	Example systems
1G	1983	Analog, FDMA / FDD	FM	voice only	AMPS, ETACS, NMT-450, NTT
2G	1991	Digital, TDMA / FDD or CDMA / FDD	GMSK, QPSK	9.6 to 14.4 kbps	GSM, IS-136 (D-AMPS), PDC, IS-95 (cdmaOne)
2.5G	1999	Digital, packet data added	GMSK, 8-PSK	57.6 to 384 kbps	HSCSD, GPRS, EDGE, IS-95B
3G	2001	Digital, W-CDMA / CDMA	QPSK, 16-QAM	144 kbps to 2 Mbps	UMTS (W-CDMA), cdma2000 1xRTT / 1xEV-DO
4G	2010	All-IP, OFDMA + MIMO	up to 64-QAM	100 Mbps mobile, 1 Gbps fixed	LTE, LTE-A, WiMAX
5G	2020	All-IP, OFDMA + massive MIMO, mmWave	up to 256-QAM	1 to 20 Gbps	5G NR

3G rates are the IMT-2000 targets: 144 kbps vehicular, 384 kbps pedestrian, 2 Mbps indoor. 4G rates are the IMT-Advanced targets. verified/added



The two upgrade paths. Top: GSM → GPRS → EDGE → W-CDMA → HSPA → LTE. Bottom: IS-95A → IS-95B → cdma2000 1x → EV-DO → LTE. **Both converge on LTE**, which is why LTE won.

2.5G protocols, with their rates

- HSCSD (TDMA path): 57.6 kbps. Still circuit switched, just more timeslots.
- GPRS (TDMA path): 171.2 kbps. First packet-switched service.
- EDGE (TDMA path): 384 kbps. Same slots, 8-PSK instead of GMSK.
- IS-95B (CDMA path): 64 kbps.

3G family, with their rates

- W-CDMA / UMTS: 2.048 Mbps. Evolution of GSM, IS-136 and PDC.
- cdma2000 1xRTT: 307 kbps. Evolution of IS-95.
- cdma2000 1xEV-DO: over 2.4 Mbps, data optimised.
- cdma2000 1xEV-DV: 144 kbps, data + voice.

What actually changed: 1G to 3G (write this, not the dates)

- **1G → 2G: analog → digital.**
 - Digital speech coding → more users in the same band.
 - Error control coding → works at lower SIR → smaller N .
 - Encryption and authentication become possible. 1G had neither.
 - SMS and low-rate data appear.
- **2G → 2.5G: circuit → packet.**
 - 2G was designed before the web → data only over a modem, 9.6 kbps.
 - GPRS adds a packet core, user pays per byte not per minute.
 - **No new radio spectrum is needed**, only new protocols → cheap upgrade.
- **2.5G → 3G: narrowband → wideband.**
 - 5 MHz W-CDMA carrier instead of 200 kHz → Mbps rates.
 - Simultaneous voice + data, video calling, multimedia.
 - Soft handoff and RAKE reception become standard (Ch 5).

What actually changed: 3G to 5G

- **3G → 4G: all-IP and OFDM.**
 - Circuit-switched core removed entirely, voice carried over IP.
 - OFDMA → immune to frequency-selective fading w/o a complex equalizer (Ch 4, Ch 5).
 - MIMO → spatial multiplexing → capacity rises w/o more spectrum.
 - Scalable bandwidth 1.4 to 20 MHz, and much lower latency.
- **4G → 5G: one service → three.**
 - eMBB (fast broadband), URLLC (1 ms latency), mMTC (a million devices per km²).
 - mmWave spectrum, massive MIMO, beamforming, network slicing.

1.4 Trends beyond 3G, and 5G

Discuss the basic trends in wireless communication beyond 3G, focusing on the key technological changes 4

HOT 4/22 82 Bh 4, 79 Bh 2, 79 Ch 2, 80 Ch 2 · 79 Bh / 79 Ch / 80 Ch ask it as "how is 5G different from 4G"

The four enabling technologies of 4G

- **OFDM / OFDMA:** splits a wideband channel into many narrow flat-fading sub-carriers → **removes the need for a complex time-domain equalizer** (Ch 4).
- **MIMO:** multiple antennas both ends → parallel spatial streams → capacity multiplies w/o extra spectrum.
- **Adaptive modulation and coding:** the modulation order follows the instantaneous SNR → near-Shannon efficiency.
- **All-IP flat architecture:** no circuit core, fewer nodes → lower latency, cheaper backhaul.

5G: what is genuinely new verified/added

- **Millimetre wave** (24 to 100 GHz): huge bandwidth available.
 - Cost: severe path loss and blockage → forces very small cells.
- **Massive MIMO:** 64 to 256 antenna elements at the BS.
 - Narrow beams steered per user → interference falls, range recovers.
- **NOMA**, non-orthogonal multiple access: several users share one resource block at different power levels, separated by successive interference cancellation (Ch 7).
- **D2D**, device-to-device: handsets talk directly, bypassing the BS.
- **Dense heterogeneous networks:** macro, micro, pico and femto cells overlaid (Ch 2).
- **C-RAN**, cloud radio access network: baseband processing centralised, only the radio head stays at the mast.
- **Network slicing:** one physical network, several virtual networks with different latency and reliability guarantees.
- **IoT / mMTC:** up to 10⁶ low-power devices per km².

Other 4G features worth listing

- Interactive multimedia, streaming video, VoIP.
- Global roaming and service portability.
- High capacity at **low cost per bit**.
- Ad hoc and multi-hop relaying.
- Better spectral efficiency (bps/Hz) than any earlier generation.

The 3 service classes (the cleanest way to say 5G Vs 4G)

- **eMBB**: enhanced mobile broadband, 1 to 20 Gbps.
- **URLLC**: ultra-reliable low latency, 1 ms, for vehicles and industry.
- **mMTC**: massive machine type communication, for IoT.
- **4G offered one pipe. 5G offers three, tuned differently.**

1.5 Comparison of available wireless systems**Compare the available wireless systems and their application areas** 4

syllabus, no direct PYQ · supports the "examples" half of the syllabus, and the cordless / cellular contrast turns up inside handoff answers (Ch 2)

System	Mobility	Coverage	TX power	Handoff	Examples
Cordless phone	very low	tens of m	mW	none	CT2, DECT
Cellular	high, vehicular	km	up to 2 W	yes, and roaming	AMPS, GSM, IS-95, LTE
Paging	high	wide area	receive only	none	POCSAG, FLEX
Cordless / PCS	pedestrian	100s of m	low	limited	PHS, PACS
hybrid					
WLAN	pedestrian, indoor	30 to 100 m	100 mW	within an ESS	IEEE 802.11 (WiFi)
WMAN	fixed to portable	km	high	yes	IEEE 802.16 (WiMAX)
Satellite	global	continental	high	spot-beam handover	Iridium, VSAT

Cordless Vs cellular (the contrast that is examinable)

- **Cordless**
 - Low mobility, low range, low speed.
 - Low power → long battery life, cheap handset.
 - **No handoff between base units**, and no roaming.
 - High circuit quality, because the link is short.
- **Cellular**
 - High mobility, wide area, vehicular speed.
 - Handoff and roaming supported → complex switching (MSC).
 - Higher transmit power (~2 W at the handset) → heavier battery.
 - Integrated with the PSTN.

Personal wireless communication (PCS / PCN)

- **PCS** = personal communication *services*: the service concept, one personal number reaching a user anywhere.
- **PCN** = personal communication *network*: the network that delivers PCS.
- In the US, "PCS" also names the **1900 MHz** band auctioned in 1994.
- Design aim: pocket handset, low power, small cells, indoor + outdoor coverage.
- **PCS is a marketing / regulatory term, not a radio technology**. Say that, then describe whichever air interface the question names.

Where the spectrum sits verified/added

- **VLF / LF**: submarine, ground wave.
- **MF / HF**: AM, short wave, ionospheric reflection.
- **VHF / UHF**: FM radio, TV, **all cellular**. Small antennas, reliable mobile links.
- **SHF**: microwave links 2 to 40 GHz, satellite C / Ku / Ka bands.
- **EHF**: mmWave, 5G.
- All of it is licensed and regulated to prevent interference (Ch 8).

1.6 Last-minute recall for this chapter

Must memorise

- **1979 NTT** first cellular, **1983 AMPS** in the USA.
- 1G analog FDMA/FDD FM, 2G digital TDMA or CDMA, 3G W-CDMA, 4G OFDMA + MIMO, 5G mmWave + massive MIMO.
- **GPRS 171.2 kbps, EDGE 384 kbps, W-CDMA 2 Mbps, LTE 100 Mbps / 1 Gbps.**
- Forward = downlink = BS → MS. Reverse = uplink = MS → BS.
- FDD = two frequencies. TDD = one frequency, two time slots.

Most repeated, in order

1. Evolution 1G → 5G **TOP 10/22**
2. Compare generations, technology advancement **TOP 9/22**
3. Forward and reverse channel **HOT 4/22**
4. Trends beyond 3G, 5G Vs 4G **HOT 4/22**
5. Duplexing and its types **PIN 1/22**

Q1 of the paper is almost always one of the first two.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELTP	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise

Short	Means	Short	Means
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK,	differential (binary) PSK
		DBPSK	
CPFSK	continuous phase FSK	MFSK,	M-ary FSK, M-ary PSK
		MPSK	
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB, VSB	double, single, vestigial sideband
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	<i>BT</i>	GMSK filter bandwidth × symbol time
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
SSMA	spread spectrum multiple access	SDMA	space division multiple access
FHMA	frequency hopped multiple access	DSMA	direct sequence multiple access (= CDMA)
FCDMA	hybrid FDMA/CDMA	DS/FHMA	hybrid direct sequence / frequency hopped
TCDMA	time division CDMA	TDFH	time division frequency hopping
IM	intermodulation (product)	MU-MIMO	multi-user MIMO
WLAN	wireless local area network	WMAN	wireless metropolitan area network
WPAN	wireless personal area network	AP	access point
BSS, IBSS	basic service set, independent (ad hoc) BSS	ESS	extended service set
CSMA/CA	carrier sense multiple access with collision avoidance	CSMA/CD	... with collision detection
DCF	distributed coordination function	PCF	point coordination function
RTS/CTS	request to send / clear to send	CCK	complementary code keying
WEP, WPA	wired equivalent privacy, WiFi protected access	U-NII	unlicensed national information infrastructure band
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
PCN	personal communication network	SMS	short message service
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet

data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

One clash to watch: **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.